INCH-POUND

MIL-C-70584B (AR)
22 August 1990
SUPERSEDING
(See 6.6)

MILITARY SPECIFICATION

CIRCUIT BOARD, CONTROL AND SYNCHRONIZER, VOLTAGE CONVERTER (PIVADS): 9397382

This specification is approved for use within the U. S. Army Armament, Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a Voltage Converter's, Control and Synchronizer (C/S) board, hereinafter referred to as "the (C/S) board," used on Gun, Anti-Aircraft Artillery, Self-Propelled (M163A2) and Towed (M167A2) Vehicles. The C/S board generates the frequency synchronization signals, initiates the start signal when the input voltage rises to the minimum voltage, and provides overvoltage protection for the voltage converter. The C/S board is a 4.56 by 5.41 inch double-sided printed circuit board mounted vertically in the voltage converter. The C/S board contains thirteen microcircuits and resistor, capacitor and semiconductor circuitry (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, U.S. Army ARDEC, ATTN: SMCAR-BAC-S, Picatinny Arsenal, New Jersey 07806-5000 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 1285

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SPECIFICATIONS

MILITARY

MIL-F-13926	-	Fire Control Materiel, Manufacture and Inspection, General Specification for
MIL-I-45607	-	Inspection Equipment, Acquisition, Maintenance and Disposition of

STANDARDS

MILITARY

MIL-STD-109	-	Quality Assurance Terms and Definitions
MIL-STD-810	-	Environmental Test Methods and Engineering Guidelines
MIL-STD-2000	-	Standard Requirements for Soldered Electrical and Electronic Assemblies

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings and publications. The following other Government documents, drawings and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DRAWINGS (See 6.4)

U. S. ARMY ARMAMENT RESEARCH, DEVELOPMENT AND ENGINEERING CENTER (ARDEC)

9397382 - Control/Synchronizer Assembly
9397408 - Control/Synchronizer Schematic

SPECIAL PACKAGING INSTRUCTION

PS9397382 - Packaging of Control/Synchronizer Assembly

(Copies of specifications, drawings, publications, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

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2.2 Order of precedence. In the event of conflict between the text of this specification and the references cited herein the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Materials. Materials shall be in accordance with the applicable drawings and specifications.

3.2 Components and assemblies. The components and assemblies shall comply with all requirements specified on Drawing 9397382, all associated drawings, and with all requirements specified in applicable specifications and standards. Where applicable, the C/S board shall meet the requirements of MIL-F-13926 as follows:

a. Dimensions and tolerances.
b. Inorganic protective surface finishings.
c. Part identification and marking.
d. Workmanship.

3.3 Environmental service conditions.

3.3.1 Vibration. The C/S board shall not suffer damage and shall meet the requirements of 3.4 and 3.5 after being subjected to continuous vibration for 3 hours along each of three mutually perpendicular axes within the frequency range of 2 to 500 hertz (Hz) and 500 to 2 hertz (Hz). The rate of change of frequency at each frequency range shall not exceed 1 Hz per second.

3.3.2 Shock. The C/S board shall not suffer damage and shall meet the requirements of 3.4 and 3.5 after exposure to a total of 18 impulse shocks consisting of three shock impulses of 15 g's ± 10%, 11 milliseconds ± 10%, half-sine wave applied in each direction of 3 mutually perpendicular axes.

3.3.3 Altitude (non-operating). The C/S board shall meet the requirements of 3.4 and 3.5 after exposure for not less than 10 minutes to barometric pressure encountered at 40,000 feet above sea level (5.5 inches of Hg).

3.3.4 Temperature.

3.3.4.1 Operating. The C/S board shall meet the requirements of 3.4 and 3.5 when stabilized thermally at any temperature between +32 deg F and +131 deg F (equipment used in a sheltered environment), -40 deg F and +155 deg F (equipment used in a field environment).

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3.3.4.2 Non-operating (storage). The C/S board shall not suffer permanent damage when subjected to the constant temperature extremes of -60 deg F for a period of 24 hours and +160 deg F for a period of 4 hours (after temperature stabilization), and shall meet the requirements of 3.4 and 3.5 when returned to room ambient.

3.3.5 Humidity (non-operating). The C/S board shall meet the requirements of 3.4 and 3.5 after exposure to an atmosphere with a relative humidity in excess of 95 percent for a period of 10 days.

3.4 Electrical. Each conductor shall be electrically continuous as specified on the schematic diagram of the C/S board.

3.5 Performance.

3.5.1 Functional description. The C/S board monitors several voltages in a voltage converter to detect abnormal voltage conditions and generates an enable signal (START) if no abnormalities are detected.

3.5.2 Supply voltages. The C/S board shall operate when 21.0 to 32.0 volts direct current (Vdc) is connected between J1-1, J1-20 (+28V) and J1-2, J1-21 (Rtn), and 14.85 to 15.15 Vdc is connected between J1-24, J1-25 (+15C) and J1-2, J1-21 (Rtn).

3.5.3 START signal. The output voltage from J1-3, J1-4, J1-22 and J1-23, with a 2500 ohm load between J1-23 and J1-2, shall be 0.1 Vdc maximum when the supply voltages specified in 3.5.2 and the following input voltages are concurrently applied:

a. 15.9 to 16.0 Vdc between J1-18, J1-36 (+15V) and J1-17, J1-35 (Rtn).

b. 25.9 to 26.0 Vdc between J1-19, J1-37 (+24V) and J1-17, J1-35 (Rtn).

c. -15.9 to -16.0 Vdc between J1-15, J1-33 (-15V) and J1-17, J1-35 (Rtn).

d. -25-9 to -26.0 Vdc between J1-16, J1-34 (-24V) and J1-17, J1-35 (Rtn).

e. 0.1 to 0.2 Vdc between J1-5 (CROW 5) and J1-14 (+5.2V Rtn).

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3.5.4 Reference voltage. The voltage at TP1 shall be 2.485 to 2.515 Vdc relative to J1-17, J1-35 (Rtn) when the supply voltages specified in 3.5.2 are applied and when each of the following voltages is individually applied:

a. 15.9 to 16.0 Vdc between J1-18, J1-36 (+15V) and J1-17, J1-35 (Rtn).

b. 25.9 to 26.0 Vdc between J1-19, J1-37 (+24V) and J1-17, J1-35 (Rtn).

3.5.5 Overvoltage protection. The output voltage from J1-3, J1-4, J1-22 and J1-23, with a 2500 ohm load between J1-23 and J1-2, shall be 8.0 to 11.0 Vdc when the supply voltages specified in 3.5.2 and the input voltages specified in paragraph 3.5.3 are applied, and when each of the following conditions is individually applied:

a. Change the voltage specified by 3.5.3a to 18.0 to 18.1 Vdc.

b. Change the voltage specified by 3.5.3b to 28.0 to 28.1 Vdc.

c. Change voltage specified by 3.5.3c to -18.0 to -18.1 Vdc.

d. Change the voltage specified by 3.5.3d to -28.0 to -28.1 Vdc.

e. Change the voltage specified by 3.5.3e to 0.8 to 1.3 Vdc, but input current shall be limited to 10 mAdc maximum.

f. Change the +28V voltage specified by 3.5.2 to 18.9 to 19.0 Vdc.

3.5.6 Synchronization signals. The C/S board shall generate the synchronization signals specified in the following when the supply voltages specified in paragraph 3.5.2 are applied:

3.5.6.1 The +5.2V SYNCH signal between J1-12 and J1-2, J-21 (Rtn) shall be a square wave having a frequency of 99.9 to 100.1 KHz, and having a low level voltage of 1.5 V maximum when sinking 12.0 mA and a high level voltage of 13.5 V minimum when sourcing 2.4 mA.

3.5.6.2 The +15V SYNCH signal between J1-13 and J1-2, J1-21 (Rtn) shall be a square wave having a frequency of 99.9 to 100.1 KHz, and having a low level voltage of 1.5 V maximum when sinking 12.0 mA and a high level voltage of 13.5 V minimum when sourcing 2.4 mA.

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3.5.6.3 The +24V SYNCH signal between J1-9 and J1-2, J1-21 (Rtn) shall be a square wave having a frequency of 99.9 to 100.1 KHz, and having a low level voltage of 1.5 V maximum when sinking 12.0 mA and a high level voltage of 13.5 V minimum when sourcing 2.4 mA.

3.5.6.4 The 115VAC SYNCH signal between J1-10 and J1-2, J1-21 (Rtn) shall be a square wave having a frequency of 49.9 to 50.1 KHz, and having a low level voltage of 1.5 V maximum when sinking 12.0 mA and a high level voltage of 13.5 V minimum when sourcing 2.4 mA.

3.5.6.5 The 800 Hz signal between J1-11 and J1-2, J1-21 (Rtn) shall be a square wave having a frequency of 799.5 to 800.5 Hz, with a low level voltage of 1.5 V maximum when sinking 12.0 mA and a high level voltage of 13.5 V minimum when sourcing 2.4 mA.

3.6 Soldering. The C/S board shall be soldered per the general requirements and Task F of MIL-STD-2000.

3.7 Burn-in. The C/S board shall meet the requirements of 3.5 after each stabilization during the following burn-in stress application: the unit shall be cycled between -40 deg C -5 deg C and 71 deg C +5 deg C at a rate of 6 deg C/minute +/-1 deg C/min with temperature stabilization at each extreme, for a total of six (6) 180-minute cycles. The unit shall be energized after stabilization at -40 deg C and deenergized after stabilization at 71 deg C. The last three (3) cycles shall be failure free.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to MIL-STD-109 to define quality assurance terms used herein.

4.1.1 Responsibility for compliance. All items shall meet all requirements of Sections 3 and 5 and all applicable drawings. The inspections set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance

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comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 General provisions. The component and subassembly inspection requirements of MIL-F-13926 form a part of the quality assurance provisions of this specification.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).
c. Special sampling inspection (see 4.5).

4.2.1 Inspection conditions. Unless otherwise specified, all examinations and tests shall be conducted at the following standard ambient conditions:

a. Temperature +50 deg F to +90 deg F
b. Relative humidity 90% RH max.
c. Pressure 28 to 32 inches of Hg

4.3 First article inspection.

4.3.1 First article submission. The contractor shall submit a first article sample as designated by the contracting officer consisting of three (3) C/S boards for inspection in accordance with 4.3.2.

4.3.2 Inspections to be performed. First article assemblies, components and test specimens may be subjected by the Government to any or all of the examinations and tests specified in Table I (Functional Test) and Table II (Special Sampling) herein and to any or all requirements of the applicable drawings.

4.3.3 Rejection. If any assembly, component or test specimen fails to comply with any of the applicable requirements, the first article sample shall be rejected. The Government reserves the right to terminate its inspection upon any failure of any assembly, component or test specimen to comply with any of the stated requirements.

4.4 Quality conformance inspection.

4.4.1 Inspection lot formation. The term "inspection lot" is defined as a homogeneous collection of units of product from which a representative sample is drawn or which is inspected 100 percent to determine conformance with applicable requirements. Units of product selected for inspection shall represent only the inspection lot from which they are drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the

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inspection lot has been produced by one manufacturer, in one unchanged process, using the same materials and methods, in accordance with the same drawings, same drawing revisions, same specifications and same specification revisions. All material submitted for inspection in accordance with this specification shall comply with the homogeneity criteria specified herein, regardless of the type of inspection procedure which is being applied to determine conformance with requirements.

4.4.2 Examinations and tests.

4.4.2.1 Components, assemblies and subassemblies. All components, assemblies and subassemblies shall be inspected in accordance with the inspection provisions contained in this specification and the Quality Assurance Provisions (QAP) listed in the Technical Data Package (TDP). In the absence of QAPs, the applicable quality assurance provisions of MIL-F-13926 shall apply.

4.4.2.2 Final acceptance inspection. Subsequent to first article approval, examinations and tests related to Section 3 herein shall be performed on a single defect (individual characteristic) basis in accordance with the sampling plans specified in 4.4.2.3 herein. Examination and test for packaging and marking shall be in accordance with Section 5 herein. The tabulated classification of defects in 4.4.2.3 shall constitute the minimum inspection to be performed by the contractor after first article approval and prior to Government acceptance or rejection by item or lot.

4.4.2.3 Functional tests. The tests in Table I shall be performed on a 100% basis.

TABLE I. Functional tests (100 percent inspection).

CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
Critical:	None	N/A
Major:		
101. Electrical	3.4	4.7.2
102. Performance	3.5	4.7.3
103. Burn-in	3.7	4.9
Minor:	None	N/A

4.5 Special sampling. One C/S board out of the first 50 produced and one out of each 100 produced thereafter shall be selected at random. The sample shall have met the requirements and tests of Table I and shall then meet the tests in Table II. Unless otherwise specified in the contract, the items shall be returned to the lot upon successful completion of testing.

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TABLE II. Special sampling.

CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
Critical:	None	N/A
Major:		
104. Vibration	3.3.1	4.7.1.1
105. Shock	3.3.2	4.7.1.2
106. Altitude	3.3.3	4.7.1.3
107. Temperature	3.3.4	4.7.1.4
108. Humidity	3.3.5	4.7.1.5
Minor:	None	N/A

4.5.1 Failure of sample. Should any one item of a special sampling fail to meet any of the specified test requirements, acceptance of the product shall be suspended by the Government until necessary corrections to all production since the last test as well as to current production has been accomplished in accordance with the provisions of the contract.

4.6 Inspection equipment.

4.6.1 Government-furnished inspection equipment. Where the contract provides for Government-furnished test equipment, supply and maintenance of test equipment shall be in accordance with the applicable requirements of MIL-I-45607.

4.6.2 Contractor-furnished inspection equipment.

4.6.2.1 Government design. Unless otherwise specified in the contract, all inspection equipment identified by drawing number in specifications or QAPs forming a part of the contract shall be supplied by the contractor in accordance with the design specified. The contractor may, however, propose alternatives to Government designs for approval in accordance with 6.3.

4.6.2.2 Contractor design. The contractor shall design and supply inspection equipment compatible with the "Methods of Inspection" specified in 4.7 of this specification and with the component inspection procedures specified in "Examination" and "Test Facilities" requirements of MIL-F-13926 whenever Government designs are not specified (see 4.6.2.1). Since tolerance of test equipment is normally considered to be within 10 percent of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions, and tolerances used in the design of test

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equipment shall be so selected and controlled as to ensure that the test equipment will reliably indicate acceptability of a product which does not exceed 90 percent of the prescribed tolerance limit and permit positive rejection when nonconforming. Construction shall be such as to facilitate routine calibration of test equipment. Contractor inspection equipment designs shall be submitted for approval in accordance with 6.3.

4.6.2.3 Test equipment. In conjunction with 4.6.2.1 and 4.6.2.2, the test equipment listed in Table IV (Test equipment list), or equivalent meeting minimum requirements, shall be used in the performance of the applicable test methods and procedures as specified in 4.7.

TABLE IV. Test equipment list.

ITEM NUMBER	DESCRIPTION
1. Power Supply #1 (+15 Vdc)	0 to 15 Vdc, adjustable. Load regulation of 1 percent or better from 0 to 100mA. Less than 100mV ripple.
2. Power Supply #2 (+28 Vdc)	0 to 32 Vdc adjustable. Load regulation of 1 percent or better from 0 to 100mA. Less than 100mV ripple.
3. Power Supply #3	0 to 30 Vdc adjustable. Load regulation of 1 percent or better from 0 to 100mA. Less than 100mV ripple. Isolated output.
4. Power Supply #4	0 to 30 Vdc adjustable. Load regulation of 1 percent or better from 0 to 100mA. Less than 1 percent ripple. Isolated output.
5. Temperature Chamber	-40 deg C to 71 deg C at a rate of ±6 deg C/minute minimum.
6. Datalogger	Multiple channel strip chart recorder; record time, date, temperatures, AC and DC voltages.
7. Digital Voltmeter	Accuracy and resolution of ±0.2 percent from 1 to 100 Vdc. Input impedance 10 megohms or greater.
8. Burn-in Load Fixture and Heatsink	Provide interface between power supply and circuit board and temperature chamber.

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TABLE IV. Test equipment list (continued).

ITEM NUMBER	DESCRIPTION
9. Ohmmeter	0 to 1000 ohms. Accuracy and resolution of 5 percent or better. Input impedance of 10K ohms minimum.
10. Oscilloscope	Input impedance of 1 megohm or greater. Dual time base with mixed sweep capabilities. Voltage ranges 0.1 to 5 Vdc per division. 20 MHz bandwidth minimum.

4.7 Methods of inspection.

4.7.1 Environmental. Environmental tests shall be conducted in accordance with the test conditions, general test performance guidance, and individual test methods of MIL-STD-810, except as specified herein. Test facilities, chambers and apparatus used in conducting the tests contained in this specification shall be capable of meeting the conditions required.

4.7.1.1 Vibration. The test shall be performed as specified in Method 514.3, Category 8, Procedure I, of MIL-STD-810 as modified by 3.3.1 herein. In Step 1 of Procedure I only steps 4 through 8 of paragraph II-2.1 apply. The C/S board shall be attached to the vibration generator directly or with a fixture, and securely held by its normal means of attachment. After vibration, the C/S board shall be tested as specified in 4.7.2 and 4.7.3 herein to determine compliance with 3.3.1.

4.7.1.2 Shock. This test shall be performed as specified in Procedure I, Section II of Method 516.3 of MIL-STD-810 except that:

a. The post test functional check (Step 5 of Procedure I) need not be conducted until completion of all shock impulses.

b. The shock pulses shall be 15 g's ± 10% amplitude and 11 ms ± 10% duration.

c. The maximum g shall occur at 5.5 milliseconds ± 10%.

d. The shock measuring system shall have a flat frequency response within ± 10% from 20 to 2,000 hertz.

After the application of all the shock impulses, the C/S board shall be tested as specified in 4.7.2 and 4.7.3 to determine compliance with 3.3.2.

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4.7.1.3 Altitude (non-operating). This test shall be performed as specified in Procedure III, Method 500.2 of MIL-STD-810. The operational checkout shall be as specified in 4.7.2 and 4.7.3 for compliance with 3.3.3.

4.7.1.4 Temperature.

4.7.1.4.1 High (operating and non-operating). This test shall be performed as specified in Procedures I and II, Method 501.2 of MIL-STD-810 with constant temperature extremes as specified in 3.3.4. The operational checkout shall be as specified in 4.7.2 and 4.7.3 for compliance with 3.3.4.

4.7.1.4.2 Low (operating and non-operating). This test shall be performed as specified in Procedures I and II, Method 502.2 of MIL-STD-810 with constant temperature extremes as specified in 3.3.4. The operational checkout shall be as specified in 4.7.2 and 4.7.3 for compliance with 3.3.4.

4.7.1.5 Humidity (non-operating). This test shall be performed in accordance with Procedure III, Method 507.2 of MIL-STD-810 except that in Step 7, checkout shall be performed at the conclusion of the test only, with the item at standard ambient conditions. The operational checkout shall be as specified in 4.7.2 and 4.7.3 for compliance with 3.3.5.

4.7.2 Electrical. Continuity shall be inspected by standard measuring equipment and visual examination for compliance with 3.4.

4.7.3 Performance. The performance of the C/S board shall be tested as follows using the equipment listed in Table III. All output parameters and voltage inputs shall comply with the requirements of 3.5.

The phrase "Power Up" shall reference the following sequence in all tests.

a. Apply power supply #1 (+15 Vdc) to pins J1-24 and RTN J1-2.

b. Apply power supply #2 (+28 Vdc) to pins J1-1 and RTN J1-2.

4.7.3.1 Oscillator frequency synchronization signals. The inverter outputs shall be tested to meet the requirements of 3.5.6 as follows:

a. Power up.

b. Verify 100 KHz $\pm$ 0.1%, square wave at pins J1-9, 12 and 13, and 50 KHz $\pm$ 0.2%, square wave at pin J1-10.

c. Verify voltage levels specified by 3.5.6.1 through 3.5.6.4.

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4.7.3.2 800 Hz. The 800 Hz output frequency shall be tested to meet the requirements of 3.5.6.5 as follows:

a. Power up.

b. Verify 800 ±0.5 Hz, square wave at pin J1-11.

c. Verify voltage levels specified by 3.5.6.5.

4.7.3.3 Reference voltage. The Reference Voltage shall be tested as follows to verify compliance with the requirements of 3.5.4 herein.

a. Power up.

b. Adjust power supply #3 to 15.9 to 16.0 Vdc, and connect between J1-18 (+15V) and J1-17 (Rtn).

c. Verify reference voltage at TP1 as required by 3.5.4.

d. Disconnect power supply #3, adjust to 25.9 and 26.0, and connect between J1-19 (+24V) and J1-17 (Rtn).

e. Verify reference voltage at TP1 as required by 3.5.4.

f. Disconnect power supplies #1, #2, and #3.

4.7.3.4 START signal and overvoltage protection. The overvoltage protection circuitry shall be tested as follows to meet the requirements of 3.5.3 and 3.5.5 herein.

a. Apply 2500 ohm load between J1-23 and J1-2, and power up. Verify START signal J1-23 (START) as required by 3.5.3. Readjust power supply #2 (28 Vdc) to 16.3 to 16.5 Vdc. Verify START signal as required by 3.5.5.

b. Turn off power supplies #1 and #2, readjust power supply #2 to 21.0 to 32.0 Vdc, and power up. Adjust power supply #3 to 15.9 to 16.0 Vdc and connect it between J1-18 (+15V) and J1-17 (Rtn). Verify START signal as required by 3.5.3. Readjust power supply #3 to 18.0 to 18.1 Vdc. Verify START signal as required by 3.5.5.

c. Turn off power supplies #1 and #2, disconnect power supply #3 and power up. Adjust power supply #3 to 25.9 to 26.0 Vdc and connect it between J1-19 (+24V) and J1-17 (Rtn). Verify START signal as required by 3.5.3. Readjust power supply #3 to 18.0 to 18.1 Vdc. Verify START signal as required by 3.5.5.

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d. Turn off power supplies #1, #2 and #3, readjust power supply #3 to 25.9 to 26.0 Vdc, power up, and turn on power supply #3. Adjust power supply #4 to -15.9 to -16.0 Vdc and connect it between J1-15 (-15V) and J1-17 (Rtn). Verify START signal as required by 3.5.3. Readjust power supply to #4 to -18.0 to -18.1 Vdc. Verify START signal as required by 3.5.5.

e. Turn off power supplies #1, #2, #3 and #4, and power up and turn on power supply #3. Adjust power supply #4 to -25.9 to -26.0 Vdc and connect it between J1-16 (-24V) and J1-17 (Rtn). Verify START signal as required by 3.5.3. Readjust power supply #4 to -28.0 to -28.1 Vdc. Verify START signal as required by 3.5.5.

f. Turn off power supplies #1, #2, #3 and #4, and power up and turn on power supply #3. Adjust power supply #4 to 0.1 to 0.2 Vdc and connect it between J1-5 (CROW5) and J1-14 (Rtn). Verify START signal as required by 3.5.3. Readjust power supply #4 to 0.8 to 1.3 Vdc with current limit set at 10 mAdc maximum. Verify START signal as required by 3.5.5.

g. Turn off power supplies #1, #2, #3 and #4.

4.8 Soldering. Soldering processes and workmanship shall be inspected by certified inspectors to the requirements and acceptance criteria of MIL-STD-2000 as specified by 3.6 herein.

4.9 Burn-in. The C/S board shall be temperature cycled and burned-in, as required by 3.7 herein, for 6 complete cycles as shown in Figure 1, for a total of 18 hours minimum, using the burn-in test setup shown in Figure 2. Temperature shall be measured at the fixture's heatsink.

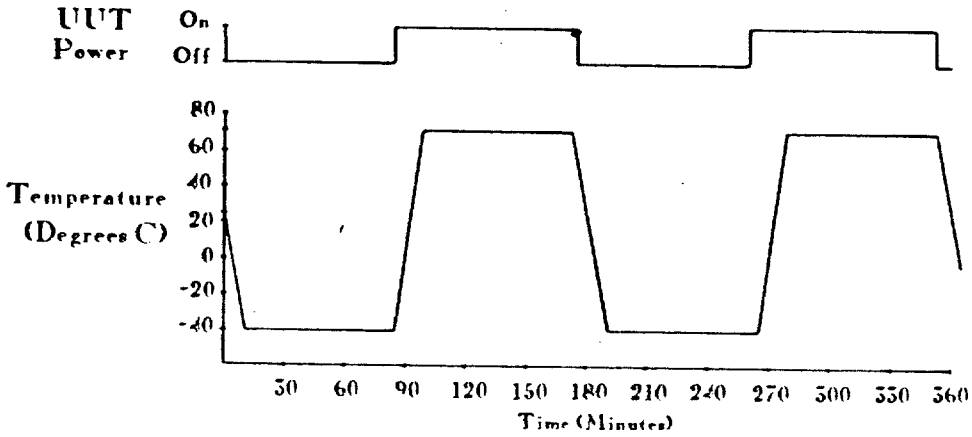


Figure 1. Burn-in Temperature Versus Time Diagram

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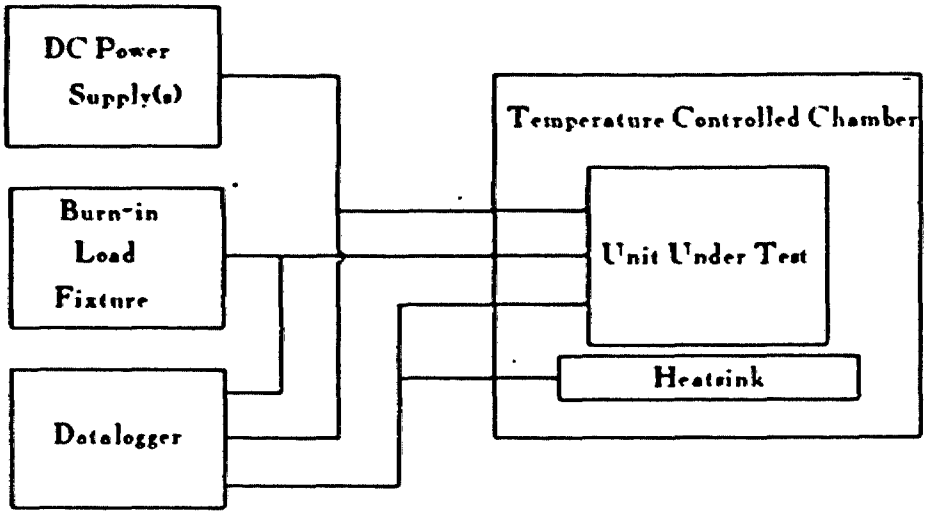


Figure 2. Burn-in Test Setup

5. PACKAGING

5.1 General. Preservation, packing and marking shall be in accordance with Special Packaging Instructions PS9397382.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The C/S board shall be used in the Voltage Converter of the PIVADS self-propelled and towed vehicles to provide input undervoltage and output overvoltage protection and synchronization and phasing signals for the dc to dc converter boards and synchronization for the dc to ac inverter board.

6.2 Acquisition requirements. Procurement documents shall specify the following:

6.2.1 Procurement requirements.

a. Title, number and date of this specification.
b. Applicable stock number.
c. First article sample requirements.
d. Packaging requirements, if other than specified in Section 5.
e. Serialization requirements, if applicable.

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6.3 Submission of contractor inspection equipment designs for approval. Submit two copies of designs as required to: Commander, U.S. Army AMCCOM, Attn: AMSMC-QAF-I, Picatinny Arsenal, NJ 07806-5000. This address will be specified on the Contract Data Requirements List, DD Form 1423 in the contract. Unless otherwise specified, data item DI-R-1714 will apply.

6.4 Drawings. Drawings listed in Section 2 of this specification under the heading U.S. Army Armament Research, Development and Engineering Center (ARDEC), may also include drawings prepared by, and identified as, U.S. Army Armament Research and Development Command (ARRADCOM), Frankford Arsenal, Rock Island Arsenal or Picatinny Arsenal drawings. Technical data originally prepared by these activities is now under cognizance of ARDEC.

6.5 Subject term (key word) listing.

Fire Control System
Gun, anti-aircraft artillery
Product Improved Vulcan Air Defense System (PIVADS)
Self-propelled 20MM M163A2
Towed 20MM M167A2
Voltage converter

6.6 Changes from previous issue. The previous issues were for ARDEC internal use only. They were not forwarded to Naval Publication and Forms Center (NPFC) for publication. This issue is initial public release for unlimited distribution.

Custodian:
Army-AR

Preparing activity:
Army-AR

(Project 1285-A746)

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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-C-70584B (AR)	22 August 1990

3. DOCUMENT TITLE

CIRCUIT BOARD, CONTROL AND SYNCHRONIZER, VOLTAGE CONVERTER (PIVADS) 9397382

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code)	
US Army ARDEC Standardization & Specification Office	(1) Commercial (201) 724-6674	(2) AUTOVON 880-6674
c. ADDRESS (Include Zip Code) ATTN: SMCAR-BAC-S Picatinny Arsenal, NJ 07806-5000	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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